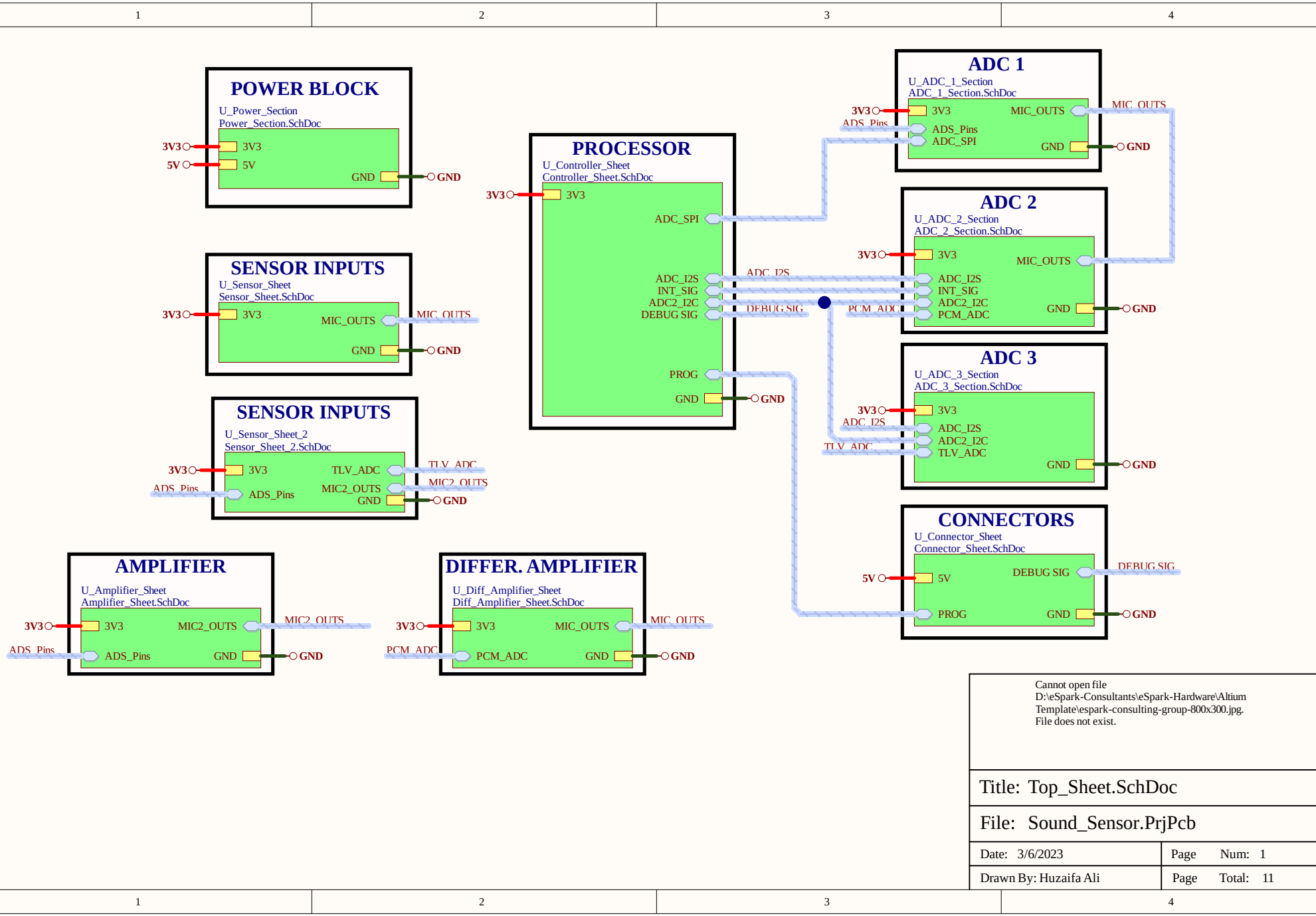
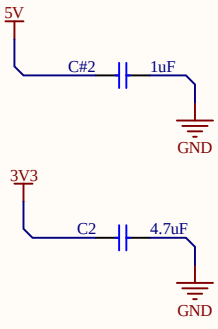


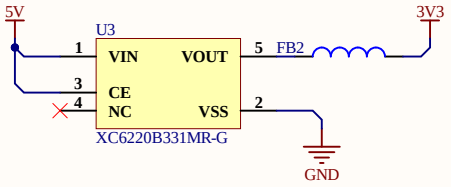
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Title: Top_Sheet.SchDoc	
File: Sound_Sensor.PrjPcb	
Date: 3/6/2023	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 11

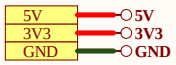
DECOUPLING CAPS



3.3V LDO



Fixed 3.3V output
150mV dropout @500mA
Max output current is 1A
Enable High to activitate the chip

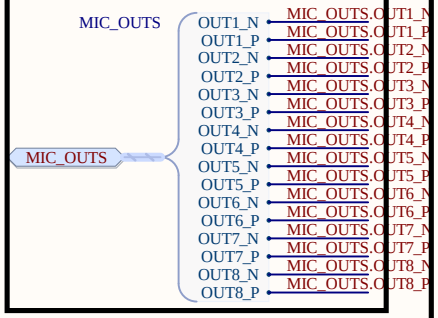


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File does not exist.

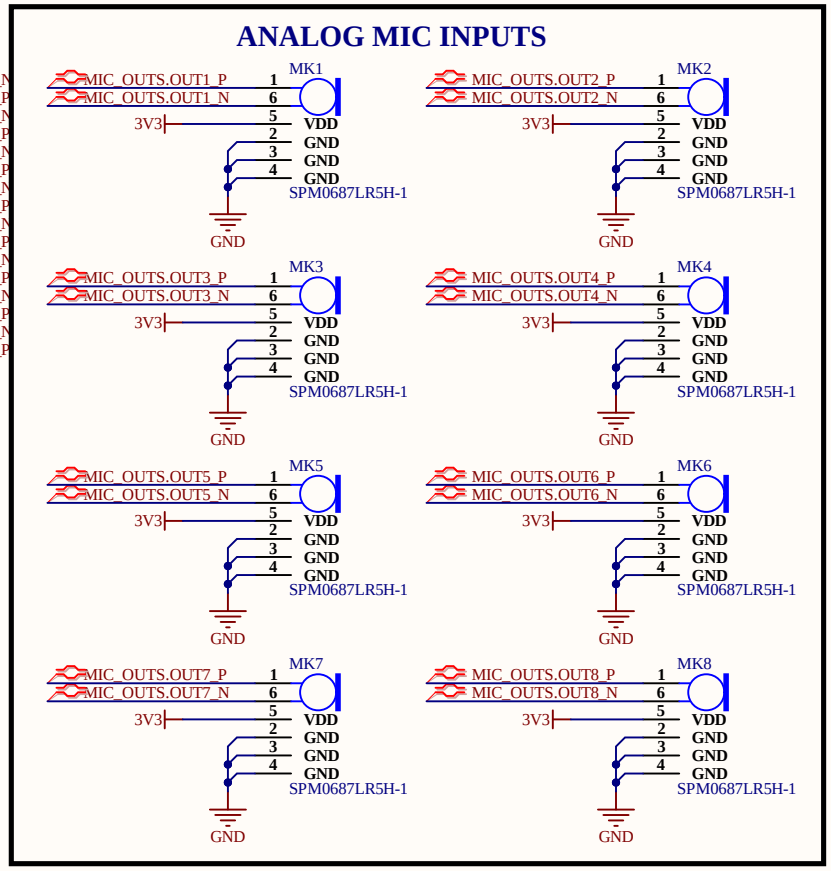
Title: Power_Section.SchDoc	
File: Sound_Sensor.PrjPcb	
Date: 3/6/2023	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 11

Set MIC differential traces to 10mils with 10mils clearnace between the two differential traces

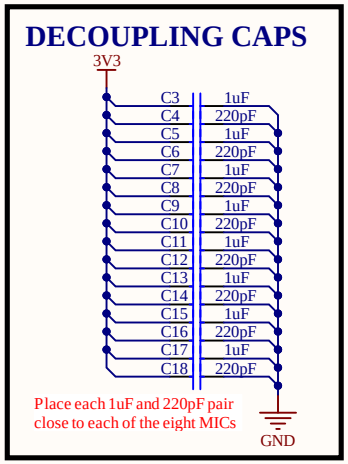
MIC OUT SIGNALS



ANALOG MIC INPUTS



DECOUPLING CAPS

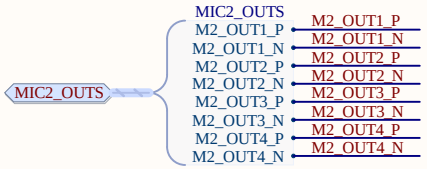




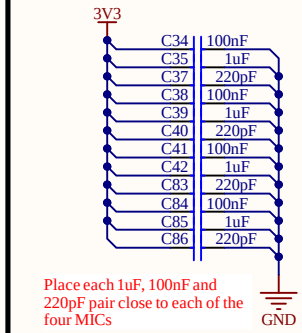
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Title: Sensor_Sheet.SchDoc	
File: Sound_Sensor.PrjPcb	
Date: 3/6/2023	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 11

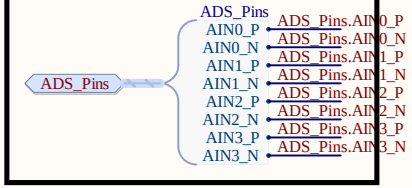
MIC2 OUT SIGNALS



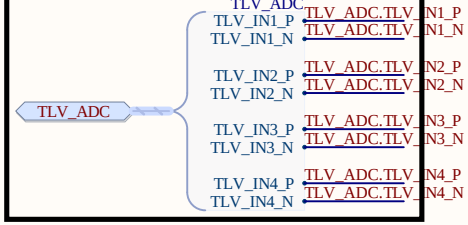
DECOUPLING CAPS



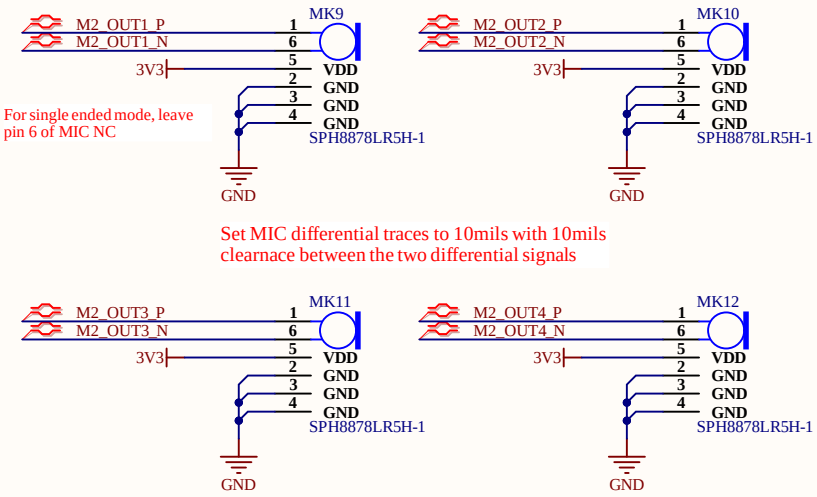
ADS131 SIGNALS



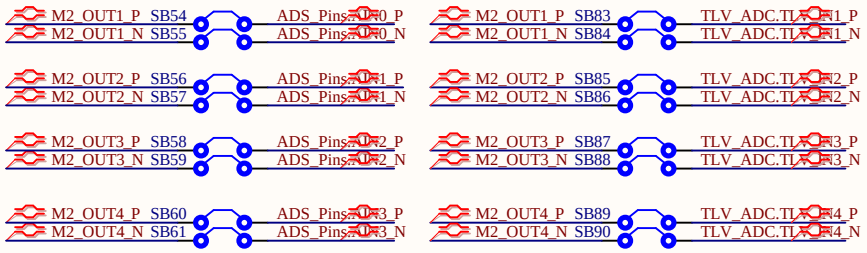
TLV320ADC5140



ANALOG MIC2 INPUTS



Selection

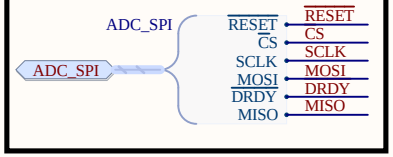




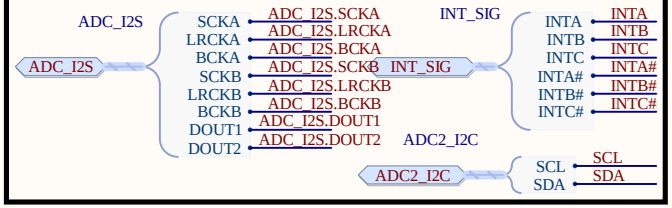
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File: Sound_Sensor.PrjPcb	
Date: 3/6/2023	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 11

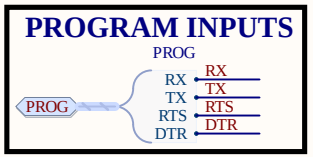
ADC1 OUTPUT



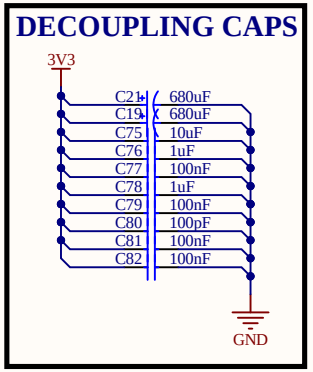
ADC 2 OUTPUT SIGNALS



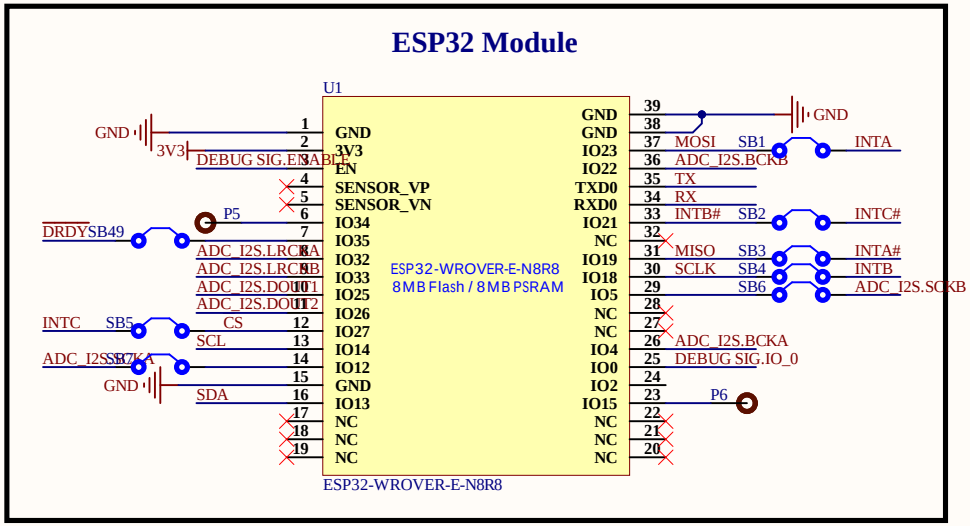
PROGRAM INPUTS



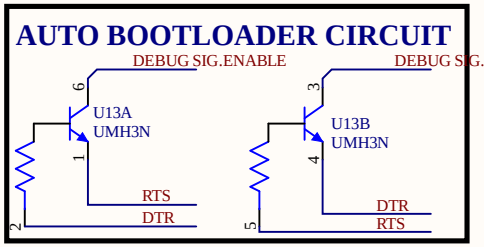
DECOUPLING CAPS



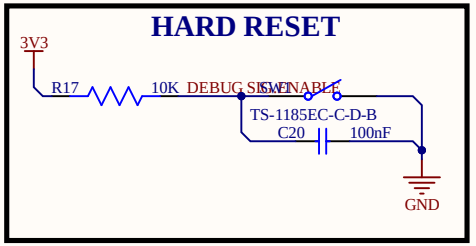
ESP32 Module



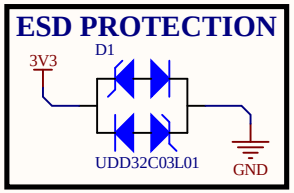
AUTO BOOTLOADER CIRCUIT

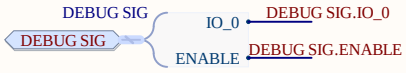


HARD RESET



ESD PROTECTION



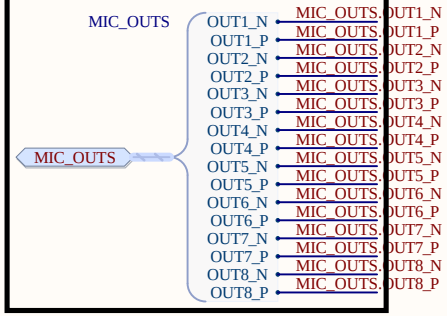




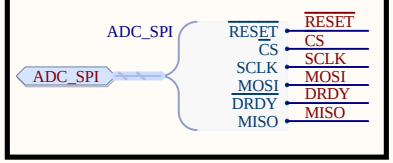
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Title: Controller_Sheet.SchDoc		
File: Sound_Sensor.PrjPcb		
Date: 3/6/2023	Page	Num: 5
Drawn By: Huzaifa Ali	Page	Total: 11

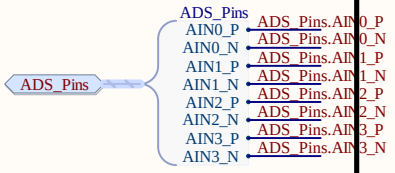
MIC OUT SIGNALS



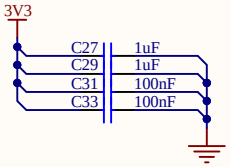
ADC OUTPUT



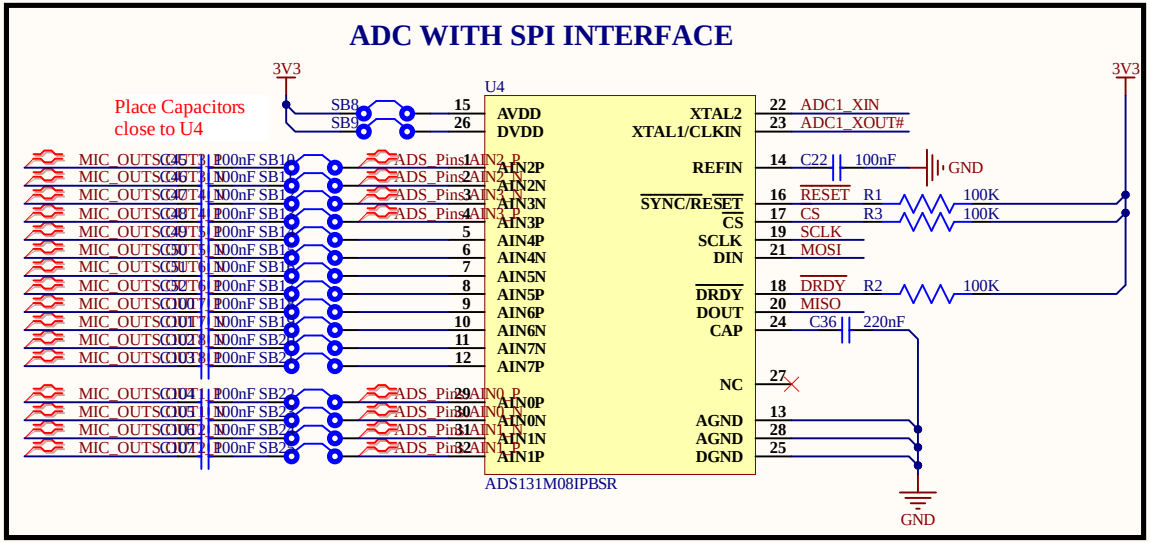
ADS131 SIGNALS



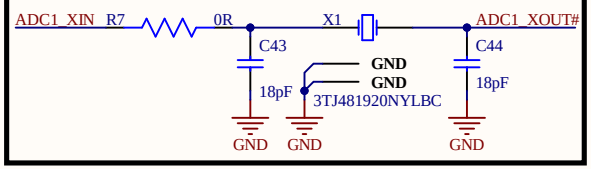
DECOUPLING CAPS



ADC WITH SPI INTERFACE



8.192 MHz Crystal





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Title: ADC_1_Section.SchDoc	
File: Sound_Sensor.PrjPcb	
Date: 3/6/2023	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 11

I2C CONTROL SIGNALS

ADC2_I2C

ADC2_I2C — SCL — SCL
SDA — SDA

MIC OUT SIGNALS

MIC_OUTS

MIC_OUTS — OUT1_N … OUT8_P — MIC_OUTS.OUT1_N … MIC_OUTS.OUT8_P

DECOUPLING CAPS

3V3
C56 10uF
C57 10uF
C58 100nF
C59 100nF
GND

24.576 MHz Crystals

XI_A_R8 0R X2 XO_A#
C61 18pF — GND GND — C62 18pF
GND GND GND

XI_B_R9 0R X3 XO_B#
C73 18pF — GND GND — C74 18pF
GND GND GND

ADC ICs

Place Capacitors close to U5

3V3
SB26 SB27 SB28 — 8 13 14
AVDD DVDD IOVDD
XO 9 XO_A#
DOUT 18 SB73 ADC_I2S.DOUT1
INT 19 INTC
VINL2/VIN1M VINR2/VIN2M VINL1/VIN1P VINR1/VIN2P VINL4/VIN4M VINR4/VIN3M VINL3/VIN4P VINR3/VIN3P
MD0 26
MD1 25
MD3 24 SCL
MD2 23 SDA
MD4 22
MD5 21 INTA
MD6 20 INTB
LDO — C53 1uF, C54 100nF — 11
Mic Bias 5
XI 10 XI_A
VREF 6 C55 1uF
SCKI 15 ADC_I2S.SCKA
LRCK 16 ADC_I2S.LRCKA
BCK 17 ADC_I2S.BCKA
AGND 7
DGND 12
GND
PCM1865DBTR

MIC_OUTS.OUT1_N … OUT8_P — C40 … C47 100nF — SB29 SB30 SB31 SB32 SB33 SB34 SB35 SB36 — PCM1_N 1, PCM2_N 2, PCM1_P 3, PCM2_P 4, PCM4_N 27, PCM3_N 28, PCM4_P 29, PCM3_P 30

Place Capacitors close to U6

3V3
SB70 SB71 SB72 — 8 13 14
AVDD DVDD IOVDD
XO 9 XO_B#
DOUT 18 ADC_I2S.DOUT2
INT 19 INTC#
VINL2/VIN1M VINR2/VIN2M VINL1/VIN1P VINR1/VIN2P VINL4/VIN4M VINR4/VIN3M VINL3/VIN4P VINR3/VIN3P
MD0 26
MD1 25
MD3 24 SCL
MD2 23 SDA
MD4 22
MD5 21 INTA#
MD6 20 INTB#
LDO — C70 1uF, C71 100nF — 11
Mic Bias 5
XI 10 XI_B
VREF 6 C72 1uF
SCKI 15 ADC_I2S.SCKB
LRCK 16 ADC_I2S.LRCKB
BCK 17 ADC_I2S.BCKB
AGND 7
DGND 12
GND
PCM1865DBTR

MIC_OUTS.OUT1_N … OUT8_P — C48 … C55 100nF — SB37 SB38 SB39 SB40 SB41 SB42 SB43 SB44 — 1 2 3 4 27 28 29 30

PCM_ADC — PCM1_N PCM2_N PCM1_P PCM2_P PCM4_N PCM3_N PCM4_P PCM3_P

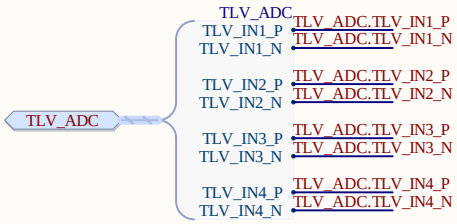
ADC_I2S — SCKA LRCKA BCKA SCKB LRCKB BCKB DOUT1 DOUT2 — ADC_I2S.SCKA … ADC_I2S.DOUT2

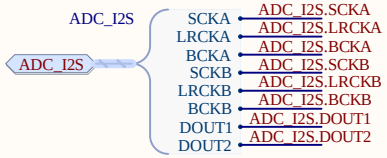
INT_SIG — INTA INTB INTC INTA# INTB# INTC#

3V3 — 3V3
GND — GND

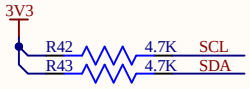
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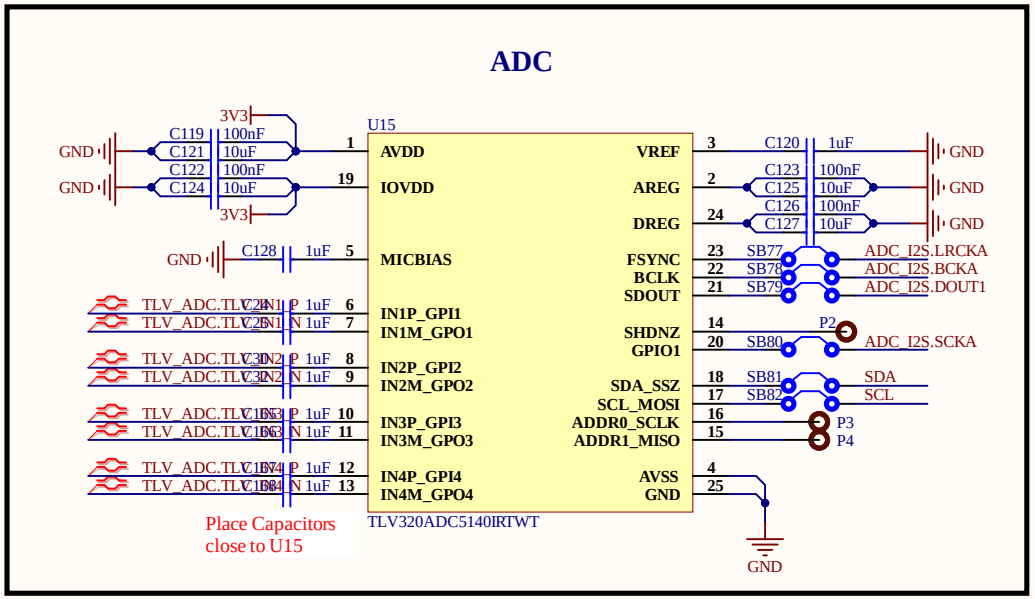
Title: ADC_2_Section.SchDoc	
File: Sound_Sensor.PrjPcb	
Date: 3/6/2023	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 11









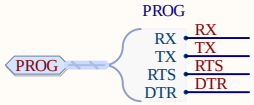




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Title: ADC_3_Section.SchDoc		
File: Sound_Sensor.PrjPcb		
Date: 3/6/2023	Page	Num: 8
Drawn By: Huzaifa Ali	Page	Total: 11

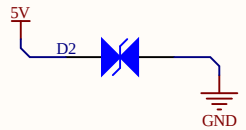
PROGRAM INPUTS



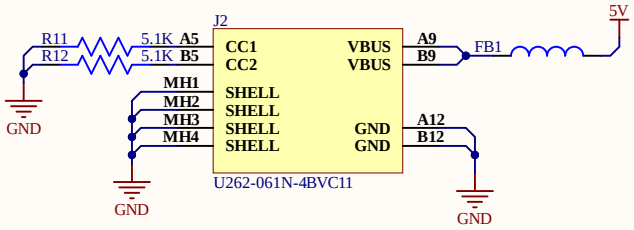
DEBUGGING SIGNALS



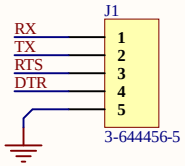
ESD PROTECTION



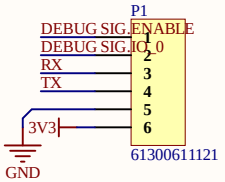
5V INPUT



PROGRAMMING HEADER



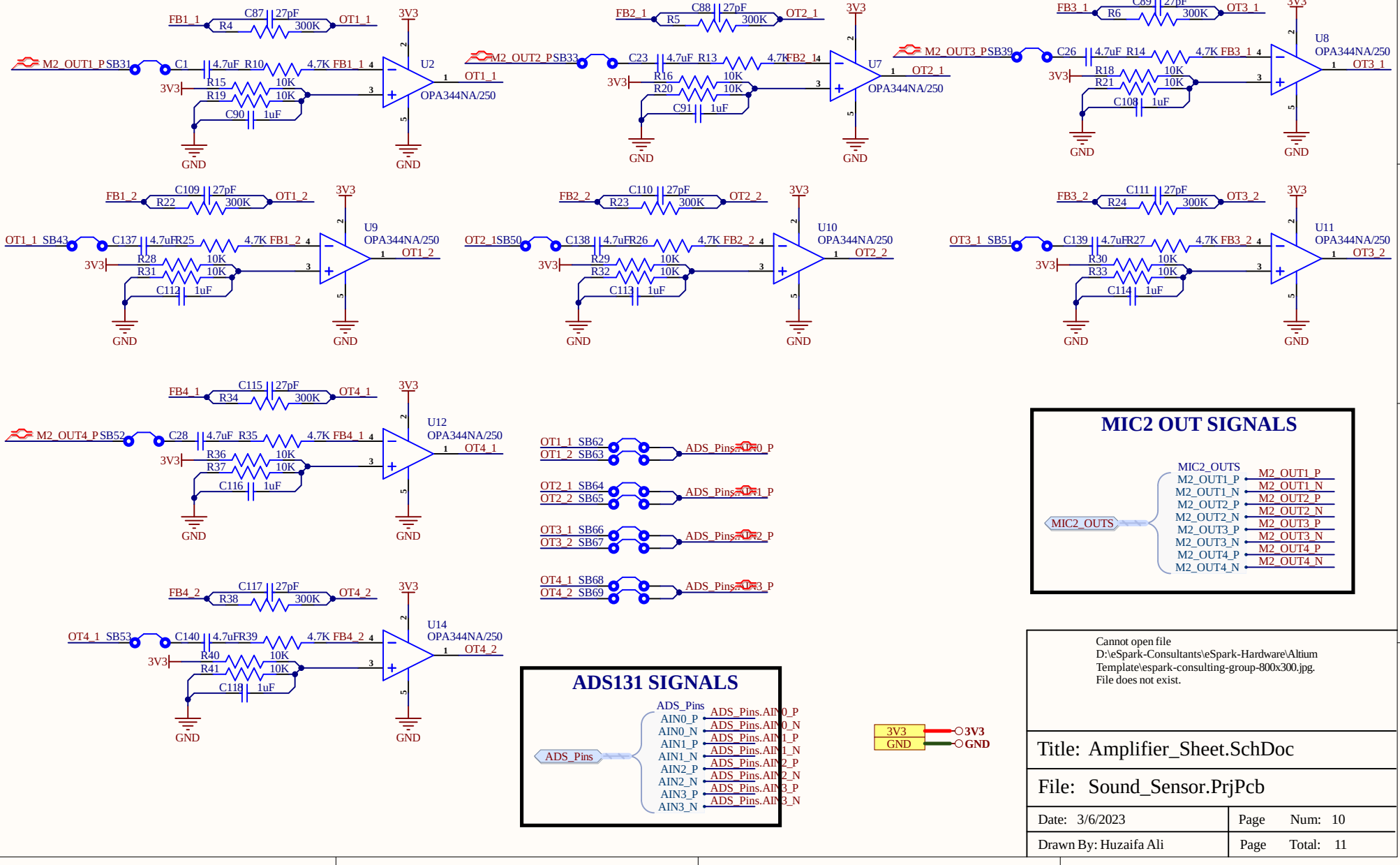
DEBUG CONNECTOR



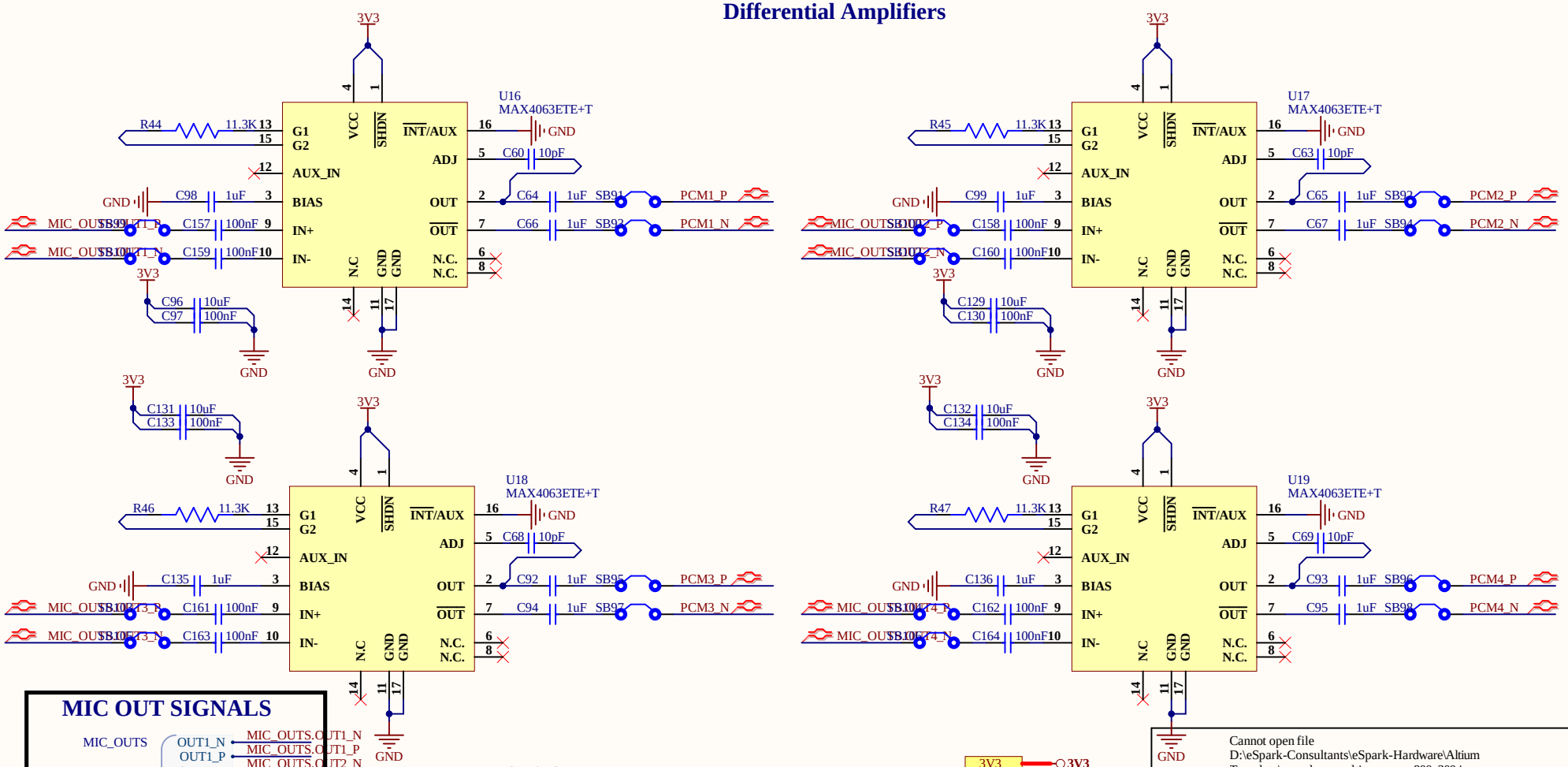


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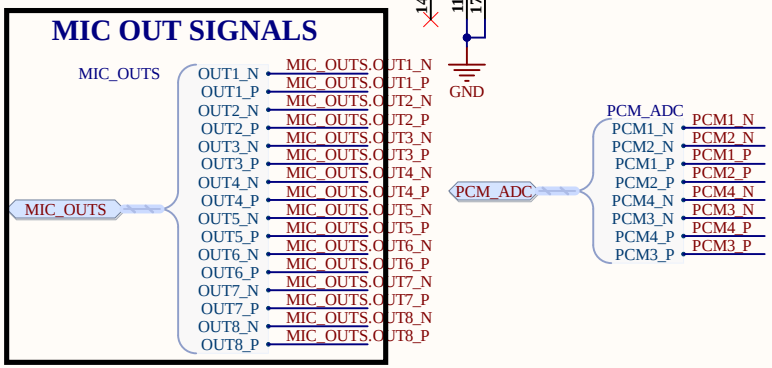
Title: Connector_Sheet.SchDoc	
File: Sound_Sensor.PrjPcb	
Date: 3/6/2023	Page Num: 9
Drawn By: Huzaifa Ali	Page Total: 11


FB1_1
C87 27pF
R4 300K
OT1_1
3V3
M2_OUT1_PSB31
C1 4.7uF
R10 4.7K
FB1_1
U2
R15 10K
R19 10K
C90 1uF
OPA344NA/250
GND
FB2_1
C88 27pF
R5 300K
OT2_1
M2_OUT2_PSB33
C23 4.7uF
R13 4.7K
FB2_1
U7
R16 10K
R20 10K
C91 1uF
FB3_1
C89 27pF
R6 300K
OT3_1
M2_OUT3_PSB39
C26 4.7uF
R14 4.7K
FB3_1
U8
R18 10K
R21 10K
C108 1uF
FB1_2
C109 27pF
R22 300K
OT1_2
OT1_1_SB43
C137 4.7uF
R25 4.7K
FB1_2
U9
R28 10K
R31 10K
C112 1uF
FB2_2
C110 27pF
R23 300K
OT2_2
OT2_1SB50
C138 4.7uF
R26 4.7K
FB2_2
U10
R29 10K
R32 10K
C113 1uF
FB3_2
C111 27pF
R24 300K
OT3_2
OT3_1_SB51
C139 4.7uF
R27 4.7K
FB3_2
U11
R30 10K
R33 10K
C114 1uF
FB4_1
C115 27pF
R34 300K
OT4_1
M2_OUT4_PSB52
C28 4.7uF
R35 4.7K
FB4_1
U12
R36 10K
R37 10K
C116 1uF
FB4_2
C117 27pF
R38 300K
OT4_2
OT4_1_SB53
C140 4.7uF
R39 4.7K
FB4_2
U14
R40 10K
R41 10K
C118 1uF
OT1_1_SB62
OT1_2_SB63
ADS_Pins.AIN0_P
OT2_1_SB64
OT2_2_SB65
ADS_Pins.AIN1_P
OT3_1_SB66
OT3_2_SB67
ADS_Pins.AIN2_P
OT4_1_SB68
OT4_2_SB69
ADS_Pins.AIN3_P
MIC2 OUT SIGNALS
MIC2_OUTS
M2_OUT1_P
M2_OUT1_N
M2_OUT2_P
M2_OUT2_N
M2_OUT3_P
M2_OUT3_N
M2_OUT4_P
M2_OUT4_N
ADS131 SIGNALS
ADS_Pins
AIN0_P
AIN0_N
AIN1_P
AIN1_N
AIN2_P
AIN2_N
AIN3_P
AIN3_N
ADS_Pins.AIN0_N
ADS_Pins.AIN1_N
ADS_Pins.AIN2_N
ADS_Pins.AIN3_N
3V3
GND
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Template\espark-consulting-group-800x300.jpg.
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File: Sound_Sensor.PrjPcb
Date: 3/6/2023
Page Num: 10
Drawn By: Huzaifa Ali
Page Total: 11

Differential Amplifiers

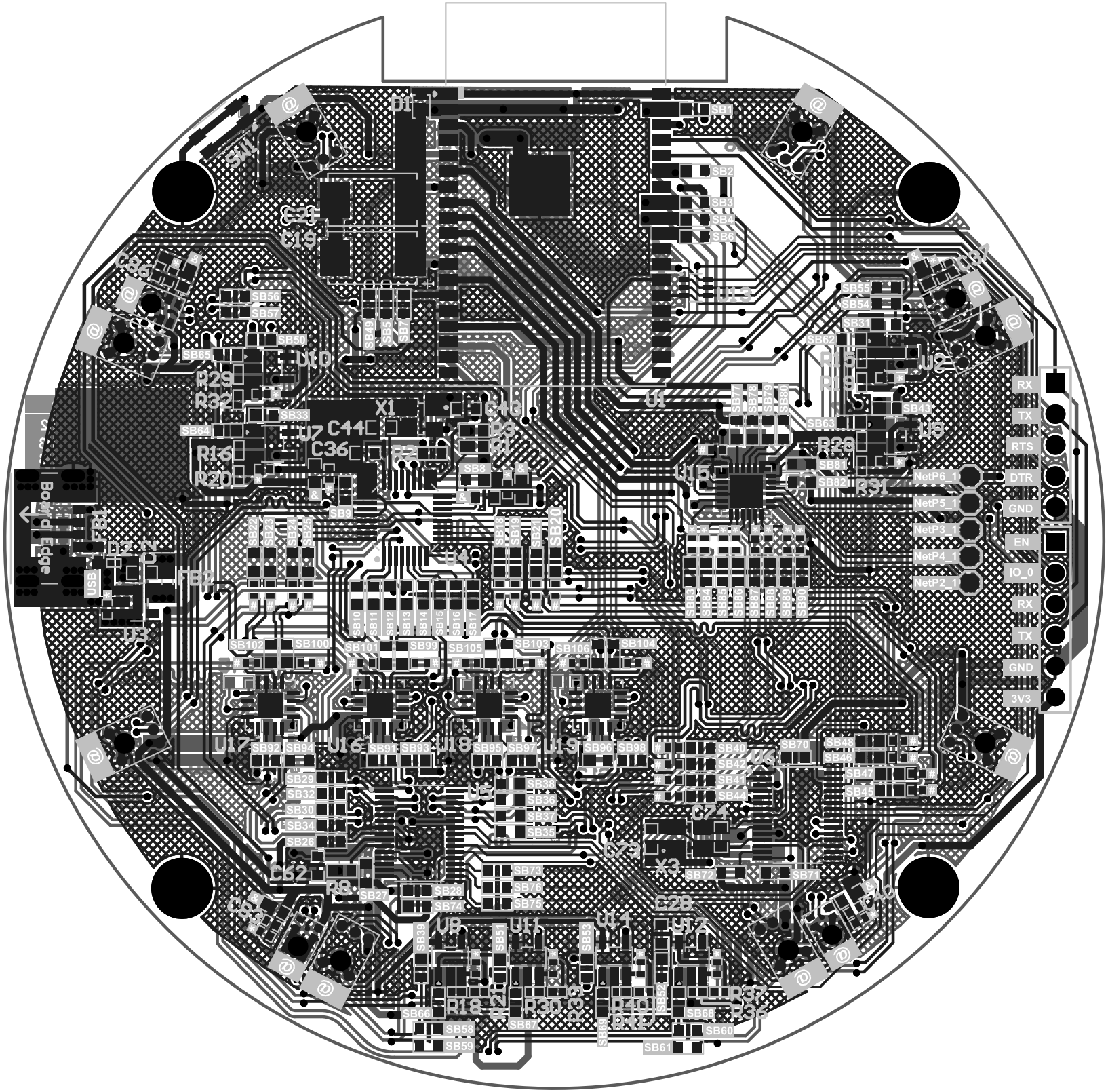


MIC OUT SIGNALS



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File: Sound_Sensor.PrjPcb	
Date: 3/6/2023	Page Num: 11
Drawn By: Huzaifa Ali	Page Total: 11


RX
TX
RTS
DTR
GND
EN
IO_0
RX
TX
GND
3V3
NetP6_1
NetP5_1
NetP3_1
NetP4_1
NetP2_1
X1
X3
U1
U2
U3
U4
U5
U6
U7
U8
U9
U10
U11
U12
U13
U14
U15
U16
U17
U18
U19
C44
C36
C19
C29
C62
C74
C28
R16
R20
R29
R32
R18
R30
R40
R37
R31
R28
D1
D2
FB1
FB2
SW1
SW2
SW3
SW4
USB
Boot
Edge